

A fish tank is 30 cm by 20 cm by 25 cm.

What is its volume?

Volume: _____

A prism has a volume of 96 cubic inches and a base area of 12 square inches.

What is its height?

Height: _____

A triangular prism has a triangle base with a base of 4 m and a height of 3 m. The prism is 7 m long.

What is its volume?

A. 84 cubic meters

B. 42 cubic meters

C. 21 cubic meters

D. 14 cubic meters

A storage box is 1.5 m by 1 m by 0.8 m.

What is its volume?

Volume: _____

True or false?

Two prisms with the same base area and the same height have the same volume.

- A.** True
- B.** False

Measure volume by counting unit cubes

MAIN	Grade 5 main question	5.MD.C.4
BACKWARD 1	Counting the squares in one layer	3.MD.C.7.a
BACKWARD 2	Equal groups	3.OA.A.1
FORWARD 1	Volume with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

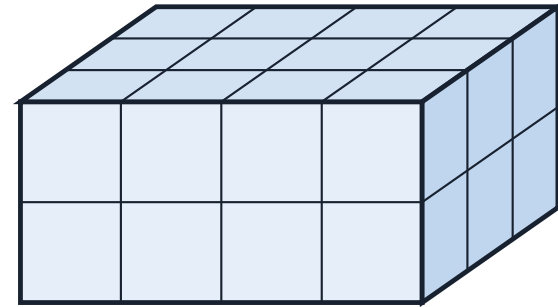
SET 47 • 5.MD.C.4

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic cm.

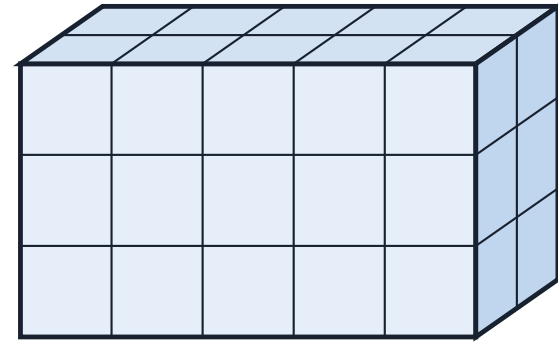
What is the volume of the prism?



Volume: _____

Each cube is 1 cubic inch.

What is the volume of the prism?

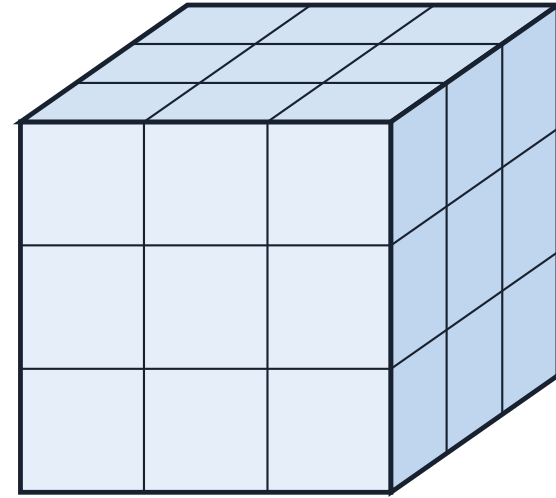


Volume: _____

Each cube is 1 cubic foot.

What is the volume of the prism?

- A.** 18 cubic feet **B.** 54 cubic feet
C. 27 cubic feet **D.** 9 cubic feet



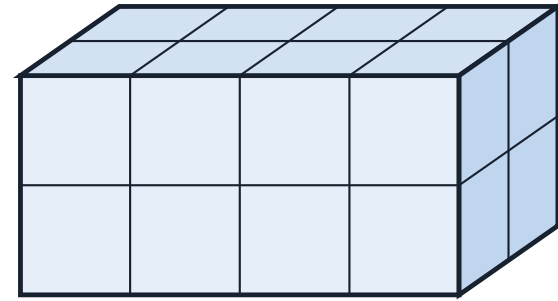
A box is filled with number cubes. The bottom layer has 6 rows of 4 cubes, and there are 3 layers.

What is the volume of the box, measured in number cubes?

Volume: _____

True or false?

Each cube is 1 cubic cm.
The volume of the prism is
16 cubic cm.



- A.** True
- B.** False

SET 47 • 5.MD.C.4

BACKWARD 1

Counting the squares in one layer • Grade 3 • Measurement & Data •
3.MD.C.7.a

The bottom layer of a box has 5 rows of 4 cubes.
How many cubes are in the layer?

Cubes: _____

True or false?

A layer with 3 rows of 6 cubes has 18 cubes.

A. True

B. False

How many unit squares cover a 4-by-6 rectangle?

A. 10

B. 46

C. 24

D. 20

A rectangle is tiled with 7 rows of 3 unit squares.
How many squares are there?

Squares: _____

True or false?

The number of unit squares that tile a rectangle is its length times its width.

- A. True
- B. False

SET 47 • 5.MD.C.4

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

There are 3 layers with 12 cubes in each layer.
How many cubes are there in all?

Cubes: _____

True or false?

4 layers of 10 cubes is 14 cubes.

A. True

B. False

Which expression shows 5 layers of 8 cubes?

A. $8 - 5$

B. 5×8

C. $5 + 8$

D. 58

Multiply.

$$6 \times 9$$

Product: _____

There are 2 layers with 15 cubes in each layer.
How many cubes are there in all?

Cubes: _____

SET 47 • 5.MD.C.4

FORWARD 1

Volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A prism is 3 cubes long, 2 cubes wide, and 2 cubes tall.
Each cube has $\frac{1}{2}$ -inch edges.

What is the volume of the prism?

Volume: _____

A box is $2\frac{1}{2}$ cm by 2 cm by 1 cm.

What is its volume?

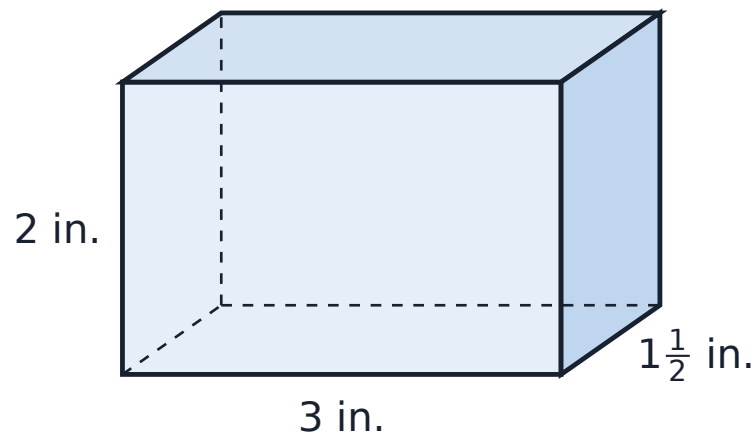
Volume: _____

How many cubes with $\frac{1}{2}$ -foot edges fill a box that is 2 ft by 1 ft by 1 ft?

- A.** 4
- B.** 8
- C.** 16
- D.** 2

Find the volume of the prism.

Volume: _____



True or false?

A cube with edges of $1\frac{1}{2}$ feet has a volume of $2\frac{1}{4}$ cubic feet.

- A.** True
- B.** False

SET 47 • 5.MD.C.4

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a rectangular base that is 5 units by 4 units.
It is 6 units tall.

What is its volume?

Volume: _____

A box has a volume of 120 cubic cm. Its base is 6 cm by 5 cm.

What is its height?

Height: _____

A prism has a base area of 15 square feet and a height of 4 feet.

What is its volume?

- A.** 60 cubic feet
- B.** 30 cubic feet
- C.** 120 cubic feet
- D.** 19 cubic feet

A triangular prism has a base area of 9 square cm and a length of 12 cm.

What is its volume?

Volume: _____

True or false?

A box that is 10 in. by 2 in. by 3 in. has a volume of 15 cubic inches.

A. True

B. False

Relate packing with unit cubes to the formulas $V = l \times w \times h$ and $V = b \times h$

MAIN	Grade 5 main question	5.MD.C.5.a
BACKWARD 1	Area by tiling	3.MD.C.7.a
BACKWARD 2	Multiplying three numbers in any grouping	3.OA.B.5
FORWARD 1	Apply the volume formulas with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms using $V = b \times h$	7.G.B.6

SET 48 • 5.MD.C.5.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.a

A prism is packed with unit cubes. It is 5 cubes long, 3 cubes wide, and 4 layers tall.

What is its volume?

Volume: _____

The bottom layer of a prism has 12 unit cubes. There are 5 layers.

Which expression gives the volume?

A. $12 \times 12 \times 5$

B. 12×5

C. 5×5

D. $12 + 5$

True or false?

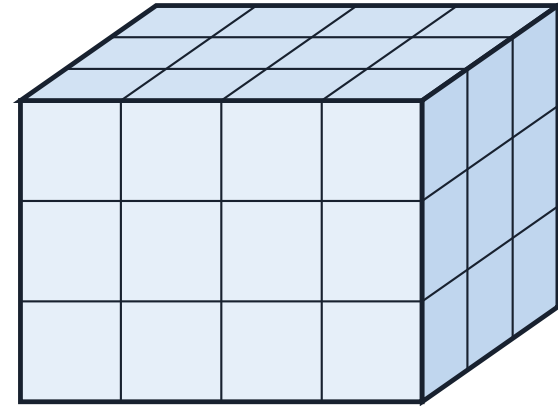
For a prism that is 4 by 3 by 2, $(4 \times 3) \times 2$ and $4 \times (3 \times 2)$ give the same volume.

A. True

B. False

Each cube is 1 cubic unit.

Find the volume by
multiplying the area of the
base by the height.



Volume: _____

A prism has a base area of 18 square units and a height of 4 units.

What is its volume?

Volume: _____

SET 48 • 5.MD.C.5.a

BACKWARD 1

Area by tiling • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is tiled with unit squares in 3 rows of 4.
What is its area?

Area: _____

True or false?

The area of a 5-by-2 rectangle is 10 square units.

A. True

B. False

Which expression gives the area of a 6-by-3 rectangle?

A. $6 - 3$

B. $6 + 3$

C. 6×3

D. $2 \times (6 + 3)$

How many unit squares tile a 9-by-2 rectangle?

Squares: _____

True or false?

Tiling a rectangle and multiplying its side lengths give the same area.

- A. True
- B. False

SET 48 • 5.MD.C.5.a

BACKWARD 2

Multiplying three numbers in any grouping • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times 3 + 4$$

A. True

B. False

Fill in the blank.

$$(5 \times 2) \times 3 = 5 \times (2 \times \underline{\quad})$$

Blank: _____

Which expression is equal to $4 \times 5 \times 2$?

A. 9×2

B. $20 + 2$

C. $4 + 5 + 2$

D. 4×10

Multiply.

$$3 \times 2 \times 5$$

Product: _____

True or false?

$$6 \times (2 \times 3) = 6 + 6$$

A. True

B. False

SET 48 • 5.MD.C.5.a

FORWARD 1

Apply the volume formulas with fractional edge lengths • Grade 6 • Geometry •
6.G.A.2

Use $V = l \times w \times h$ to find V when $l = 3\frac{1}{2}$ in., $w = 2$ in.,
and $h = 1\frac{1}{2}$ in.

Volume: _____

Use $V = b \times h$ to find the volume of a prism with $b = 6\frac{1}{4}$ square cm and $h = 4$ cm.

Volume: _____

A prism is packed with $\frac{1}{2}$ -unit cubes: 6 cubes long, 4 cubes wide, and 2 cubes tall.

Which shows its volume?

A. $6 + 4 + 2 = 12$ cubic units

B. $3 \times 2 \times 1 = 6$ cubic units

C. $6 \times 4 \times 2 = 48$ cubic units

D. $\frac{1}{2} \times 48 = 24$ cubic units

A cube has edges that are $\frac{3}{4}$ foot long.

What is its volume?

Volume: _____

True or false?

For a rectangular prism, $V = l \times w \times h$ and $V = b \times h$ give the same volume.

- A. True
- B. False

SET 48 • 5.MD.C.5.a

FORWARD 2

Volume of right prisms using $V = b \times h$ • Grade 7 • Geometry • 7.G.B.6

A triangular prism has a triangle base with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

A prism has a trapezoid base with an area of 14 square inches. The prism is 5 inches tall.

What is its volume?

Volume: _____

A prism has a volume of 84 cubic meters and a base area of 12 square meters.

What is its height?

A. 1,008 m

B. 7 m

C. 72 m

D. 96 m

A rectangular prism is 8 by 5 by 2.5.

What is its volume?

Volume: _____

True or false?

$V = b \times h$ works for triangular prisms as well as rectangular prisms.

A. True

B. False

Apply $V = l \times w \times h$ and $V = b \times h$ to solve real-world problems

MAIN

Grade 5 main question

5.MD.C.5.b

BACKWARD 1

Multiplying multi-digit numbers

4.NBT.B.5

BACKWARD 2

The area of the base

4.MD.A.3

FORWARD 1

Real-world volume with fractional edge lengths

6.G.A.2

FORWARD 2

Real-world problems with volume and surface area

7.G.B.6

SET 49 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A fish tank is 24 inches long, 12 inches wide, and 15 inches tall.

What is its volume?

Volume: _____

A sandbox has a base area of 30 square feet. It is filled with sand to a depth of 2 feet.

How much sand does it hold?

Volume: _____

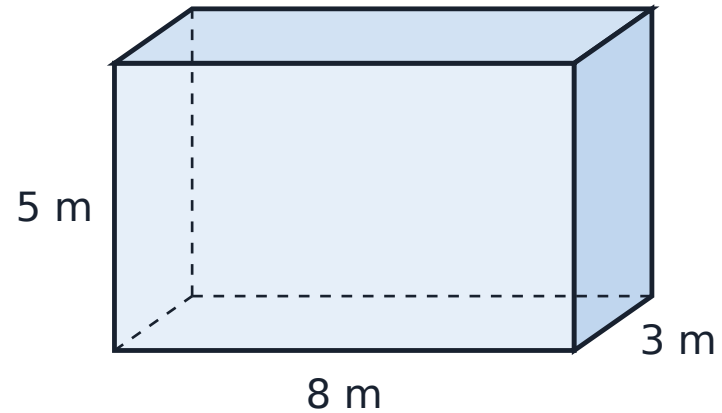
A box is 9 cm long, 4 cm wide, and 5 cm tall.

What is its volume?

- A.** 90 cubic cm
- B.** 36 cubic cm
- C.** 180 cubic cm
- D.** 18 cubic cm

Find the volume of the prism.

Volume: _____



A cube-shaped box has edges that are 7 inches long.
What is its volume?

Volume: _____

SET 49 • 5.MD.C.5.b

BACKWARD 1

Multiplying multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$24 \times 12$$

Product: _____

Multiply.

$$288 \times 5$$

Product: _____

True or false?

$$36 \times 5 = 150$$

A. True

B. False

$$49 \times 7 = ?$$

- A.** 56
- B.** 343
- C.** 283
- D.** 353

Multiply.

$$15 \times 24$$

Product: _____

SET 49 • 5.MD.C.5.b

BACKWARD 2

The area of the base • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 24 in. by 12 in.

What is its area?

Area: _____

True or false?

A 9-by-4 rectangle has an area of 36 square units.

A. True

B. False

The base of a box is 8 m by 3 m.

What is the area of the base?

- A.** 11 square meters
- B.** 24 square meters
- C.** 22 square meters
- D.** 48 square meters

A square has sides that are 7 inches long.
What is its area?

Area: _____

A rectangle has an area of 30 square feet and a length of 6 feet.

What is its width?

Width: _____

SET 49 • 5.MD.C.5.b

FORWARD 1

Real-world volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $1\frac{1}{2}$ ft by 1 ft by $\frac{3}{4}$ ft.

What is its volume?

Volume: _____

A prism has a base area of $12\frac{1}{2}$ square inches and a height of 3 inches.

What is its volume?

Volume: _____

A cube has edges that are $2\frac{1}{2}$ cm long.

What is its volume?

A. $6\frac{1}{4}$ cubic cm

B. 15 cubic cm

C. $7\frac{1}{2}$ cubic cm

D. $15\frac{5}{8}$ cubic cm

A drawer is 2 ft by $1\frac{1}{2}$ ft by $\frac{1}{2}$ ft.

What is its volume?

Volume: _____

True or false?

A box that is 3 by $\frac{1}{2}$ by 4 has a volume of 12 cubic units.

A. True

B. False

SET 49 • 5.MD.C.5.b

FORWARD 2

Real-world problems with volume and surface area • Grade 7 • Geometry •
7.G.B.6

A tent is a triangular prism. Its triangle has a base of 2 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

A pool is 10 m long, 5 m wide, and 2 m deep. It is filled to $\frac{3}{4}$ of its depth.

How much water is in the pool?

Volume: _____

A box has a volume of 360 cubic inches. Its base is 12 in. by 6 in.

What is its height?

A. 3 inches

B. 30 inches

C. 5 inches

D. 60 inches

A rectangular prism is 4.5 cm by 2 cm by 3 cm.
What is its surface area?

Surface area: _____

True or false?

A triangular prism with a base area of 8 square feet and a length of 5 feet holds 40 cubic feet.

A. True

B. False

Find a missing dimension when the volume is known

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Finding an unknown factor	3.OA.A.4
FORWARD 1	Solve one-step equations of the form $px = q$	6.EE.B.7
FORWARD 2	Solve two-step equations	7.EE.B.4.a

SET 50 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 96 cubic inches. It is 6 inches long and 4 inches wide.

How tall is it?

Height: _____

A prism has a volume of 150 cubic cm and a height of 6 cm.

What is the area of its base?

Base area: _____

A tank holds 240 cubic feet. Its base is 8 feet by 5 feet.

How deep is the tank?

- A.** 6 feet
- B.** 30 feet
- C.** 40 feet
- D.** 227 feet

A rectangular prism has a volume of 84 cubic meters. It is 7 m long and 4 m tall.

How wide is it?

Width: _____

True or false?

A box with a volume of 60 cubic units and a base area of 12 square units is 5 units tall.

A. True

B. False

SET 50 • 5.MD.C.5.b

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$96 \div 4$$

Quotient: _____

Divide.

$$150 \div 6$$

Quotient: _____

True or false?

$$240 \div 8 = 3$$

A. True

B. False

$$84 \div 7 = ?$$

A. 12

B. 77

C. 11

D. 13

Divide.

$$60 \div 5$$

Quotient: _____

SET 50 • 5.MD.C.5.b

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$6 \times \square = 48$$

Number: _____

True or false?

If $\square \times 5 = 35$, then $\square = 7$.

A. True

B. False

What number goes in the box?

$$4 \times \square = 36$$

A. 9

B. 8

C. 32

D. 40

What number goes in the box?

$$\square \times 7 = 63$$

Number: _____

True or false?

If $8 \times \square = 56$, then $\square = 6$.

A. True

B. False

SET 50 • 5.MD.C.5.b

FORWARD 1

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for h.

$$24h = 96$$

$$h = \underline{\hspace{10cm}}$$

Solve for b.

$$6b = 150$$

$$b = \underline{\hspace{10cm}}$$

Solve for w.

$$28w = 84$$

- A.** 56
- B.** 2,352
- C.** 112
- D.** 3

A prism has a volume of 52.5 cubic cm and a base area of 15 square cm.

Write and solve an equation to find its height h .

Equation and solution: _____

True or false?

The solution of $\frac{3}{4}x = 12$ is $x = 9$.

A. True

B. False

SET 50 • 5.MD.C.5.b

FORWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A box is 4 in. wide, 5 in. tall, and $(x + 2)$ in. long. Its volume is 120 cubic inches.

Find x .

$x =$ _____

Solve for x.

$$6(x + 3) = 54$$

x = _____

Solve for h.

$$12h + 8 = 68$$

A. 5

B. 6.3

C. 60

D. 4

Two boxes have a total volume of 100 cubic feet. One box holds 40 cubic feet. The other has a base area of 12 square feet.

Write and solve an equation to find the height h of the other box.

Equation and solution: _____

True or false?

The solution of $3(x - 2) = 21$ is $x = 9$.

A. True

B. False

Find the volume of a figure made of two rectangular prisms

MAIN	Grade 5 main question	5.MD.C.5.c
BACKWARD 1	Area of figures made of rectangles	3.MD.C.7.d
BACKWARD 2	Volume of one rectangular prism	5.MD.C.5.b
FORWARD 1	Volume of composite figures with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of composite solids	7.G.B.6

SET 51 • 5.MD.C.5.c

MAIN

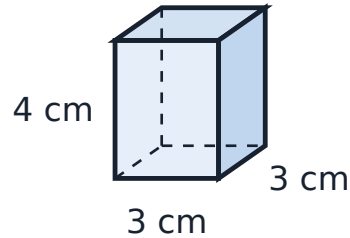
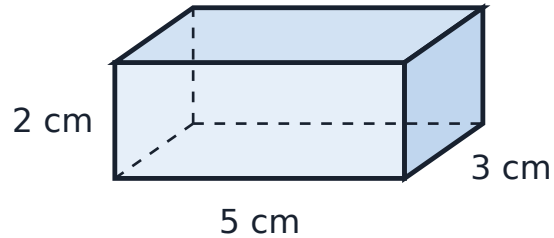
Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.c

A figure is made of two rectangular prisms that do not overlap. One is 6 ft by 4 ft by 3 ft. The other is 2 ft by 4 ft by 5 ft.

What is the total volume?

Volume: _____

A figure is made of the two prisms shown, which do not overlap.
What is the total volume?



Volume: _____

An L-shaped step is made of a 4-by-2-by-1 prism with a 2-by-2-by-1 prism on top.

What is its volume?

- A.** 12 cubic units
- B.** 8 cubic units
- C.** 10 cubic units
- D.** 16 cubic units

A building is a 10 m by 8 m by 6 m prism with a 4 m by 8 m by 3 m prism attached to one side.

What is the total volume?

Volume: _____

True or false?

The volume of a figure made of two prisms that do not overlap is the sum of their volumes.

- A. True
- B. False

SET 51 • 5.MD.C.5.c

BACKWARD 1

Area of figures made of rectangles • Grade 3 • Measurement & Data •
3.MD.C.7.d

A figure is made of a 5-by-3 rectangle and a 2-by-4 rectangle that do not overlap.

What is its area?

Area: _____

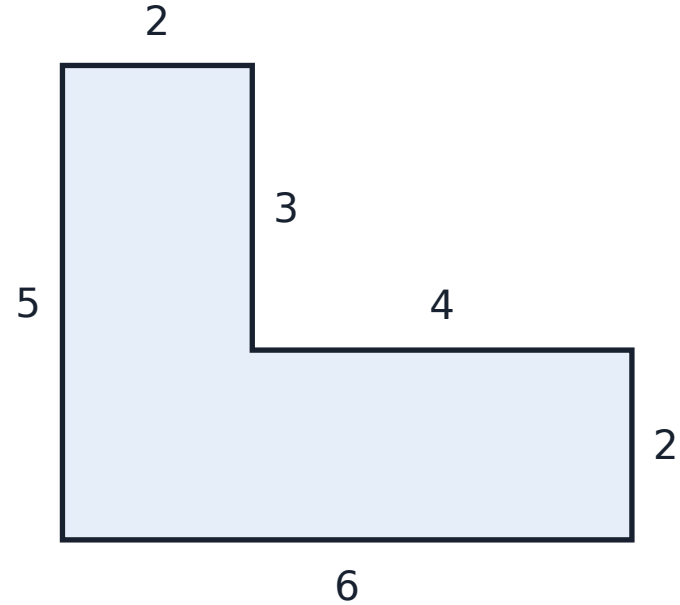
True or false?

The area of a figure made of two non-overlapping parts is the sum of the areas of the parts.

- A.** True
- B.** False

What is the area of the figure?

- A.** 18 square units
- B.** 22 square units
- C.** 12 square units
- D.** 30 square units



A figure is made of two squares that do not overlap.
Their sides are 3 units and 5 units long.

What is the total area?

Area: _____

A 2-by-2 square is cut out of an 8-by-6 rectangle.
What is the area of the part that is left?

Area: _____

SET 51 • 5.MD.C.5.c

BACKWARD 2

Volume of one rectangular prism • Grade 5 • Measurement & Data •
5.MD.C.5.b

What is the volume of a prism that is 6 by 4 by 3?

Volume: _____

What is the volume of a prism that is 2 by 4 by 5?

Volume: _____

True or false?

A prism with a base area of 32 square units and a height of 3 units has a volume of 35 cubic units.

- A. True
- B. False

What is the volume of a prism that is 4 by 8 by 3?

- A.** 96 cubic units
- B.** 15 cubic units
- C.** 120 cubic units
- D.** 32 cubic units

A cube has edges that are 5 units long.

What is its volume?

Volume: _____

SET 51 • 5.MD.C.5.c

FORWARD 1

Volume of composite figures with fractional edge lengths • Grade 6 • Geometry
• 6.G.A.2

A figure is made of two prisms that do not overlap: $2\frac{1}{2}$ ft by 2 ft by 1 ft, and 1 ft by 2 ft by $\frac{1}{2}$ ft.

What is the total volume?

Volume: _____

Two prisms are joined. One has a volume of $4\frac{1}{2}$ cubic inches. The other is 3 in. by 1 in. by $\frac{1}{2}$ in.

What is the total volume?

Volume: _____

A step is made of a 3-by-1-by- $\frac{1}{2}$ prism and a 1-by-1-by- $\frac{1}{2}$ prism that do not overlap.

What is its volume?

A. 2 cubic units

B. $\frac{1}{2}$ cubic unit

C. 4 cubic units

D. $1\frac{1}{2}$ cubic units

A block that is 1 in. by 1 in. by $2\frac{1}{2}$ in. is cut out of a box that is 4 in. by 3 in. by $2\frac{1}{2}$ in.

What volume is left?

Volume: _____

True or false?

A figure made of two cubes with $\frac{1}{2}$ -foot edges has a volume of 1 cubic foot.

- A. True
- B. False

SET 51 • 5.MD.C.5.c

FORWARD 2

Volume of composite solids • Grade 7 • Geometry • 7.G.B.6

A figure is a 6-by-4-by-3 rectangular prism with a triangular prism on top. The triangle has a base of 4 and a height of 2, and the triangular prism is 6 long.

What is the total volume?

Volume: _____

A concrete step is made of two prisms: 1.5 m by 0.5 m by 0.2 m and 1.5 m by 0.25 m by 0.4 m.

What is the total volume?

Volume: _____

A 10-by-10-by-10 cube has a 2-by-2-by-10 hole cut all the way through it.

What volume remains?

- A.** 960 cubic units
- B.** 800 cubic units
- C.** 1,000 cubic units
- D.** 40 cubic units

A shed is a 3 m by 4 m by 2.5 m prism with a roof shaped like a triangular prism. The triangle has a base of 3 m and a height of 1 m, and the roof is 4 m long.

What is the total volume?

Volume: _____

True or false?

The volume of a composite solid can be found by adding the volumes of its parts.

A. True

B. False

Understand the coordinate plane: axes, origin, and coordinates

MAIN	Grade 5 main question	5.G.A.1
BACKWARD 1	Whole numbers on a number line	2.MD.B.6
BACKWARD 2	Perpendicular lines	4.G.A.1
FORWARD 1	Name the coordinates of points in all four quadrants	6.NS.C.6.c
FORWARD 2	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d

SET 52 • 5.G.A.1

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.1

What is the point where the x-axis and the y-axis meet called?

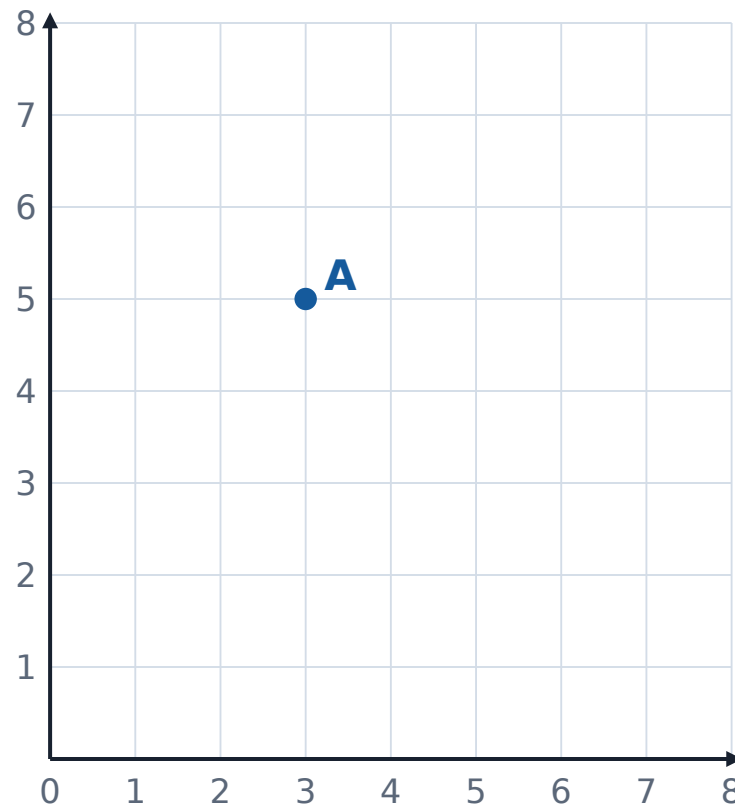
- A.** The y-axis
- B.** The origin
- C.** The x-coordinate
- D.** A quadrant

In the ordered pair $(7, 2)$, what does the 7 tell you?

Answer: _____

What are the coordinates of point A?

A = _____



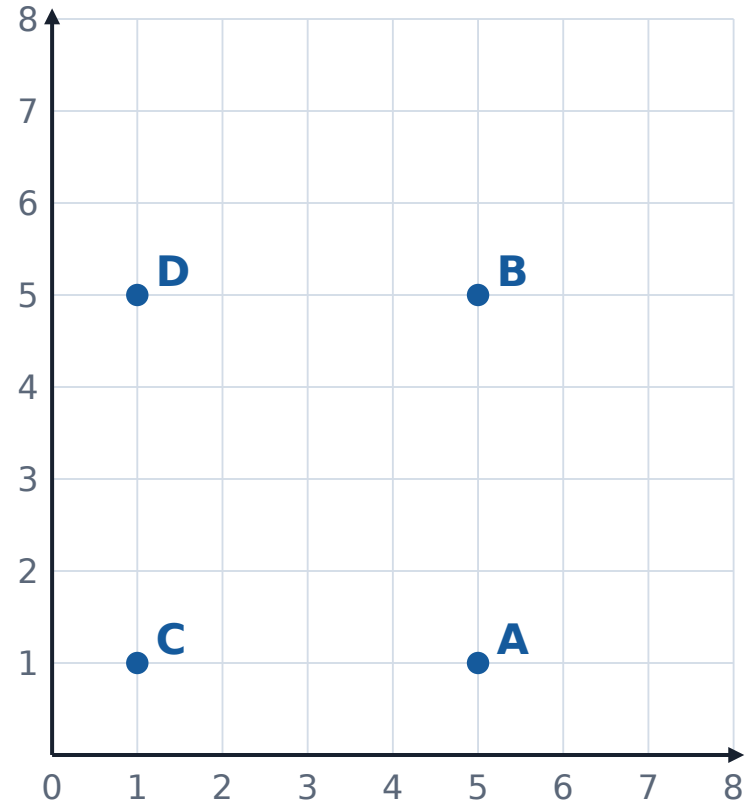
True or false?

To reach $(0, 6)$, you start at the origin and move 6 units to the right along the x-axis.

- A. True
- B. False

Which point is at (5, 1)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



SET 52 • 5.G.A.1

BACKWARD 1

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point P?



P = _____

True or false?

On a number line, numbers increase from left to right.

- A.** True
- B.** False

Which number is 4 units to the right of 0?

- A.** 4
- B.** 8
- C.** 40
- D.** 0

How many units is it from 0 to 9 on a number line?

Units: _____

True or false?

On a number line, 3 is farther from 0 than 7 is.

A. True

B. False

SET 52 • 5.G.A.1

BACKWARD 2

Perpendicular lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

A. True

B. False

The x-axis and the y-axis are ____.

- A.** perpendicular
- B.** parallel
- C.** curved
- D.** the same line

How many degrees are in a right angle?

Degrees: _____

True or false?

Parallel lines never meet.

A. True

B. False

Which letter is made of two perpendicular line segments?

A. S

B. L

C. O

D. C

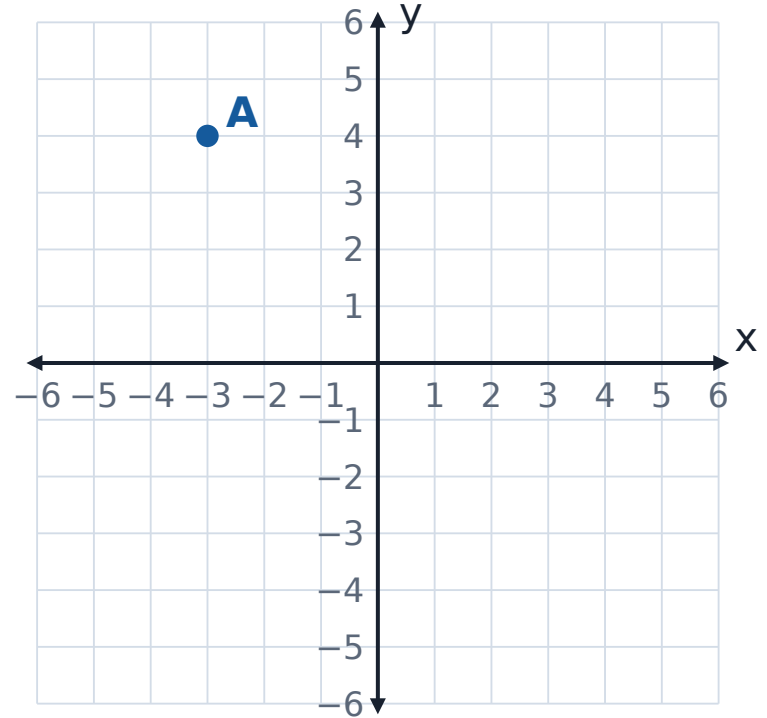
SET 52 • 5.G.A.1

FORWARD 1

Name the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

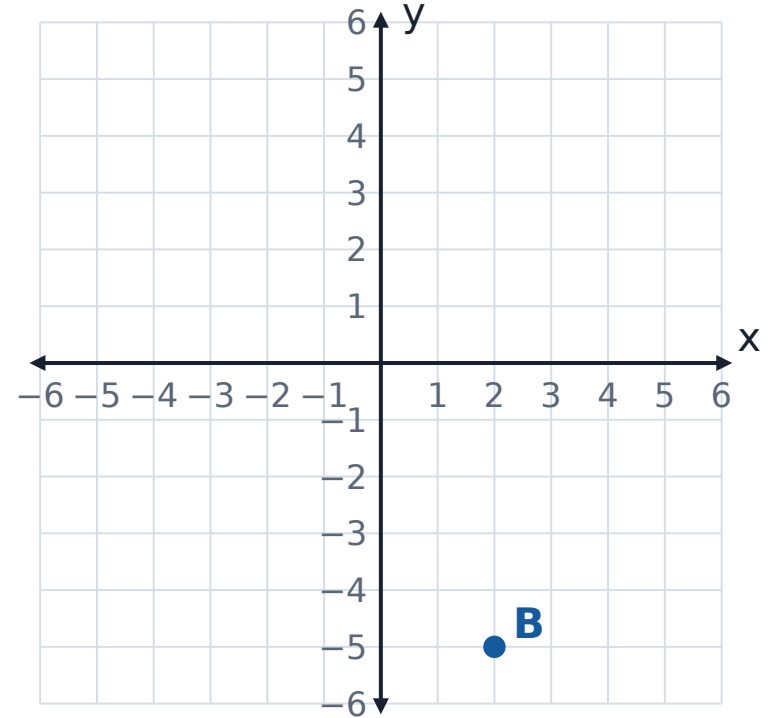
What are the coordinates of point A?

A = _____



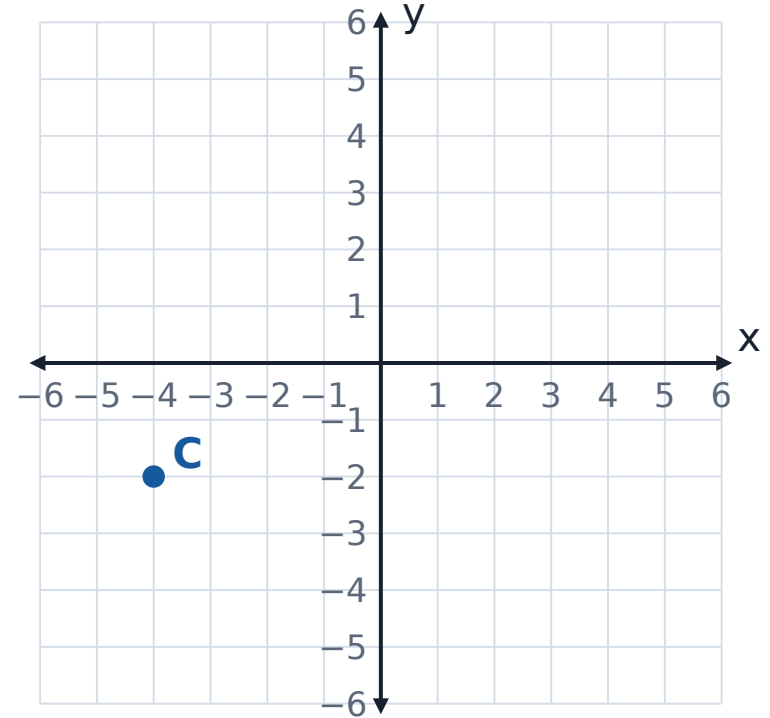
What are the coordinates of point B?

B = _____



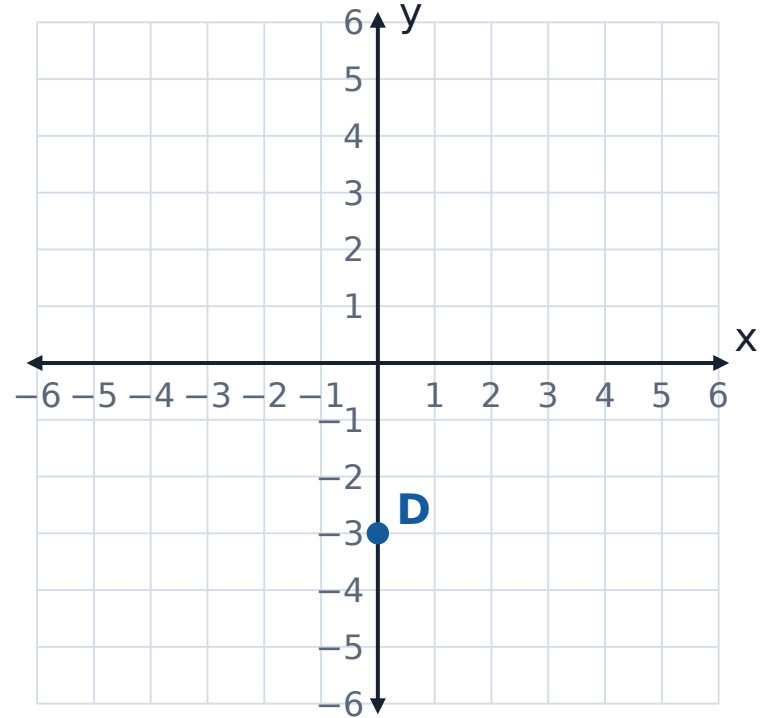
Which ordered pair names point C?

- A.** $(-4, -2)$
- B.** $(-2, -4)$
- C.** $(4, 2)$
- D.** $(-4, 2)$



What are the coordinates of point D?

D = _____

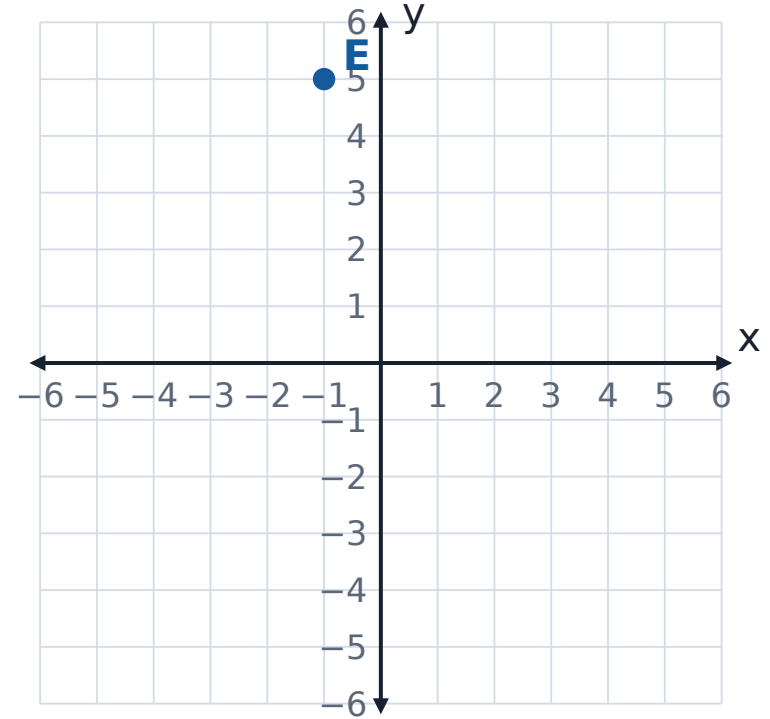


True or false?

Point E is located at $(5, -1)$.

A. True

B. False



SET 52 • 5.G.A.1

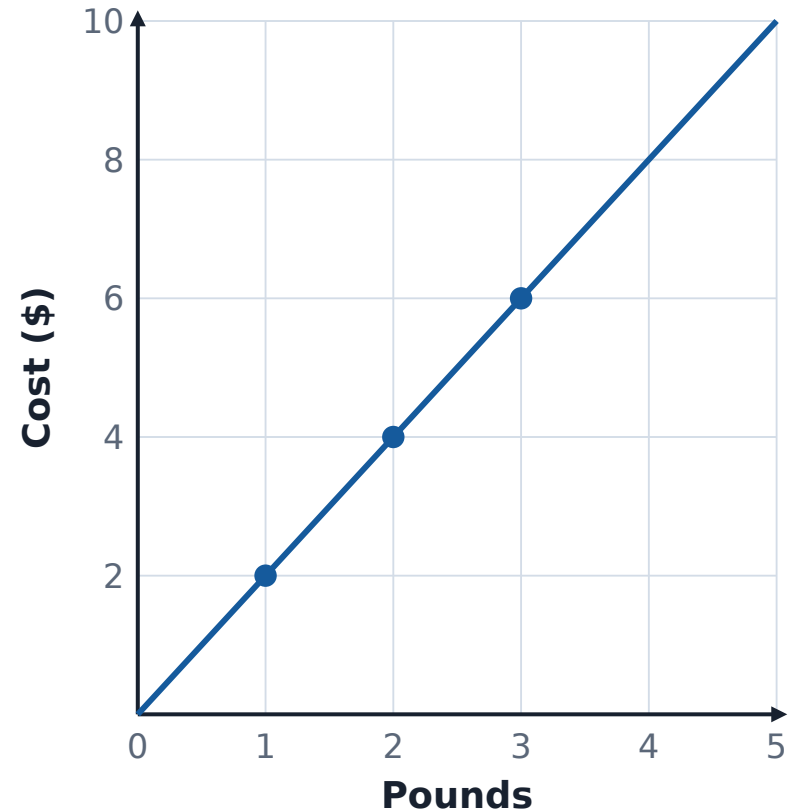
FORWARD 2

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of pears.

What does the point (3, 6) represent?

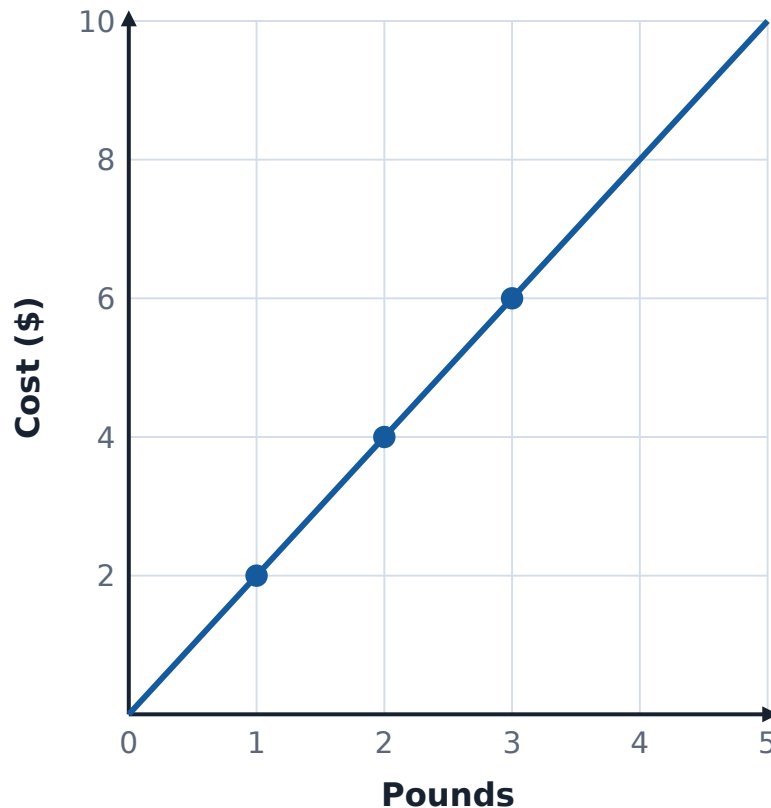
Meaning: _____



The graph shows the cost of pears.

What does the point (1, 2) represent?

- A.** The cost starts at \$2.
- B.** 2 pounds cost \$1.
- C.** \$1 buys 2 pounds.
- D.** 1 pound costs \$2.



True or false?

On the graph of a proportional relationship between pounds and cost, the point $(0, 0)$ means 0 pounds cost \$0.

- A.** True
- B.** False

The graph of a proportional relationship passes through $(1, r)$ and $(4, 20)$.

What is r ?

$r =$ _____

A graph shows the distance y in miles after x hours. It passes through $(2, 90)$.

What does this point mean?

- A.** The speed is 2 miles per hour.
- B.** In 90 hours, the distance is 2 miles.
- C.** The speed is 90 miles per hour.
- D.** In 2 hours, the distance is 90 miles.

Graph points in the first quadrant

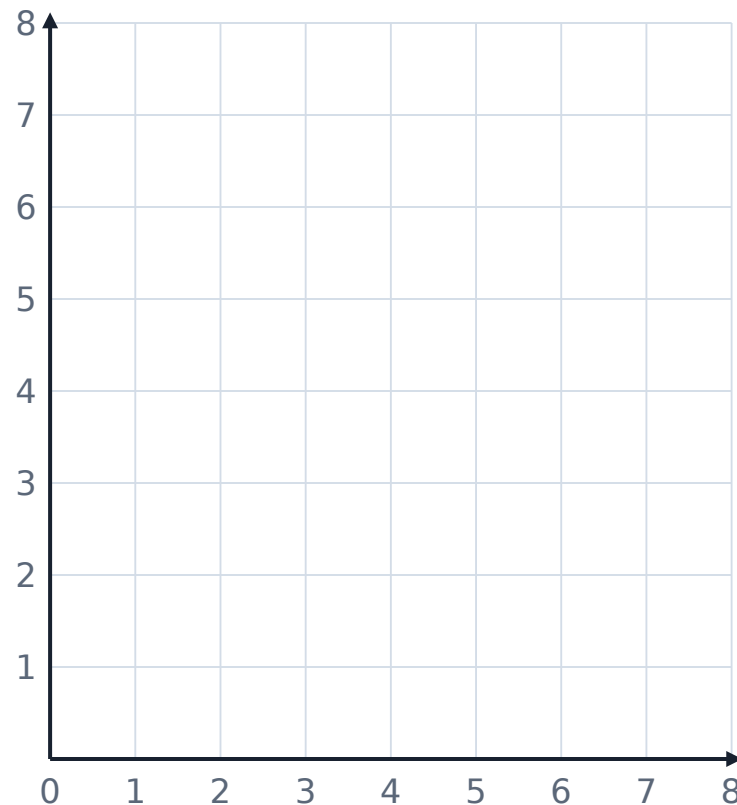
MAIN	Grade 5 main question	5.G.A.2
BACKWARD 1	Reading coordinates	5.G.A.1
BACKWARD 2	Reading the scale on an axis	3.MD.B.3
FORWARD 1	Plot and reflect points in all four quadrants	6.NS.C.6.b
FORWARD 2	Graph pairs to decide whether a relationship is proportional	7.RP.A.2.a

SET 53 • 5.G.A.2

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

Plot and label the points $A(2, 7)$, $B(6, 3)$, and $C(0, 5)$.

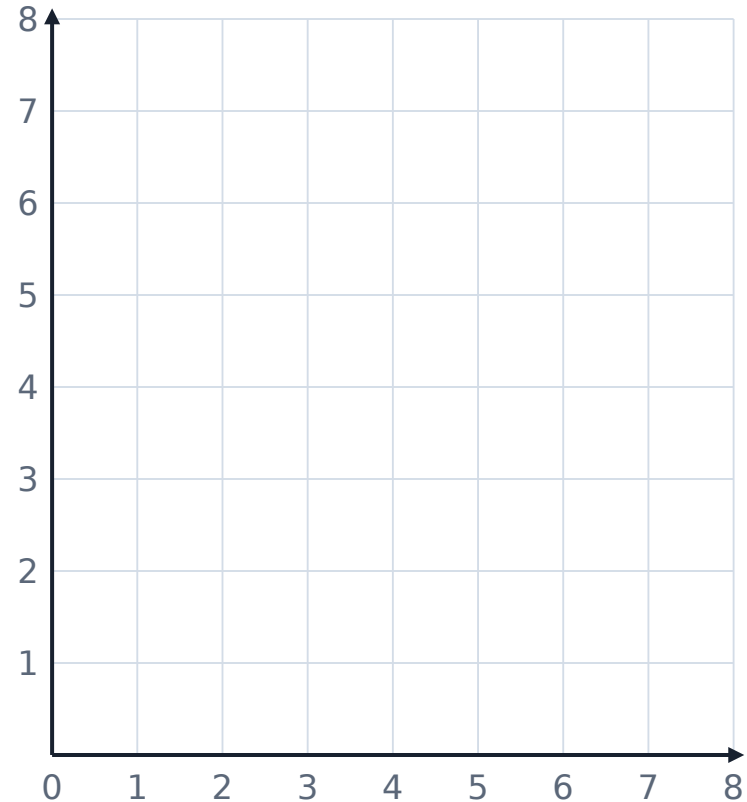


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

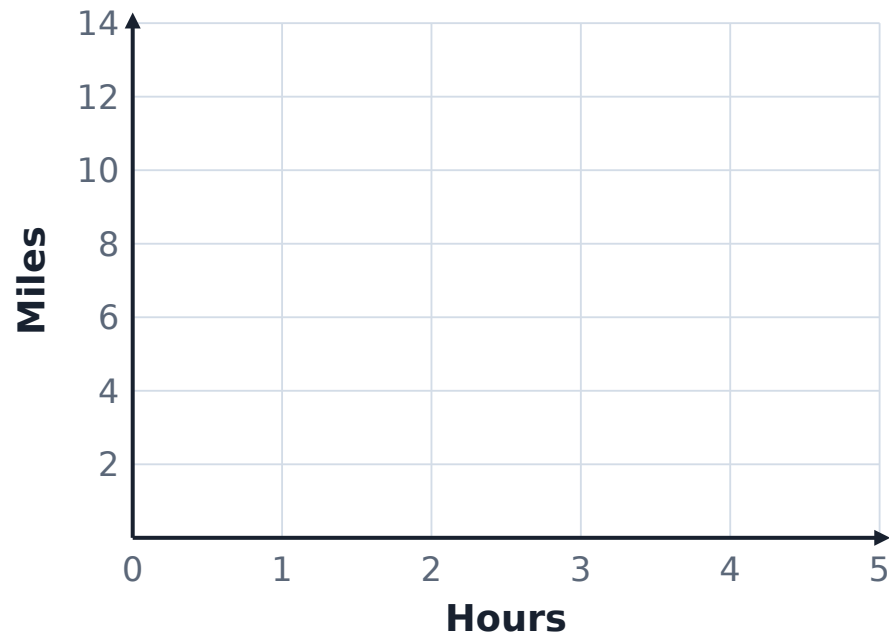
Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q1/5

Plot and label the points D(4, 4), E(7, 0), and F(1, 6).

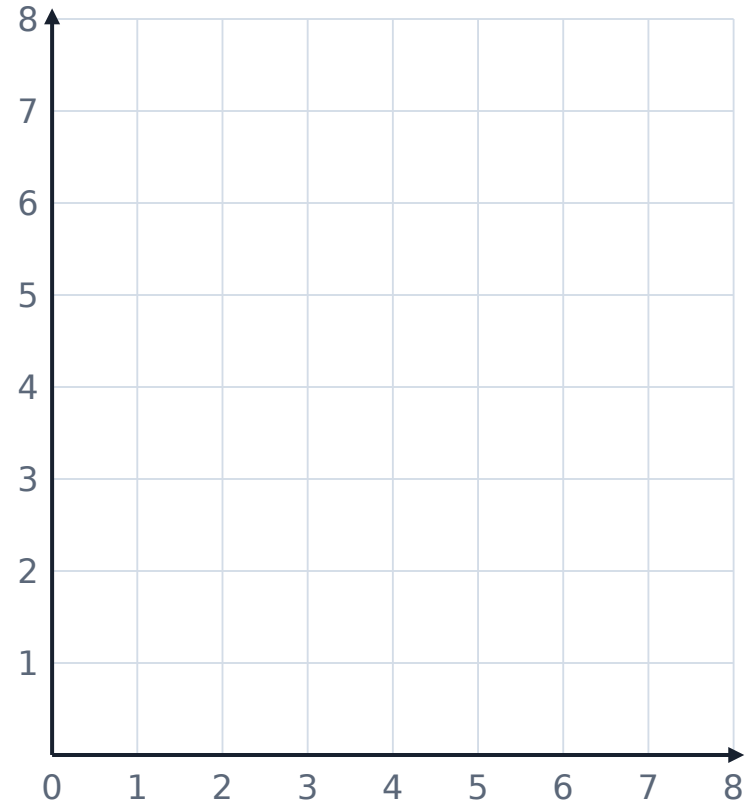


Plot the points from the table.

Hours	1	2	3	4
Miles	3	6	9	12



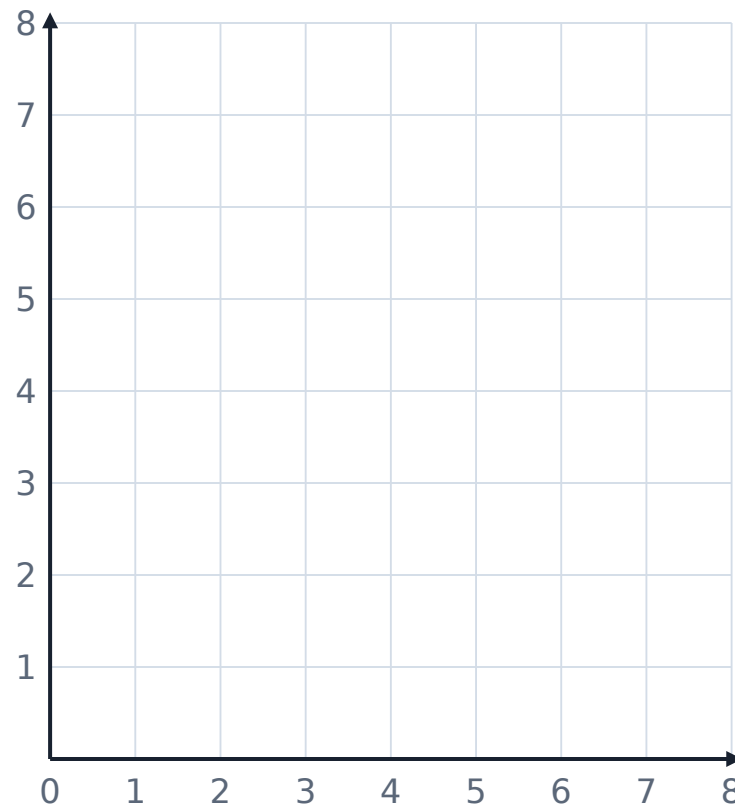
Plot and label the points $G(3, 2\frac{1}{2})$ and $H(5\frac{1}{2}, 4)$.



Plot $P(2, 3)$ and $Q(2, 7)$.

How far apart are the two points?

Distance: _____



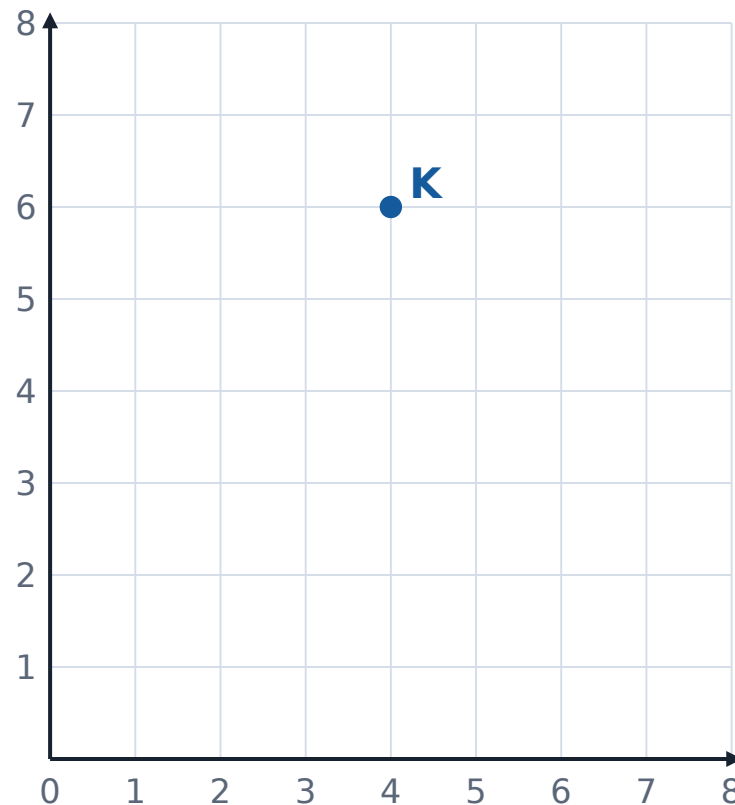
SET 53 • 5.G.A.2

BACKWARD 1

Reading coordinates • Grade 5 • Geometry • 5.G.A.1

What are the coordinates of point K?

K = _____



True or false?

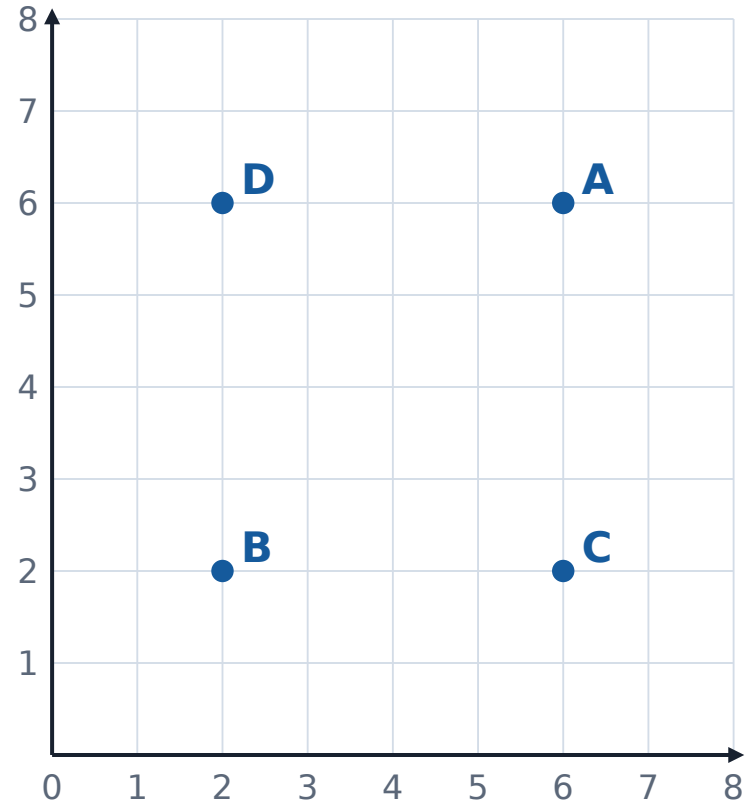
To plot (3, 8), you move up 3 units.

A. True

B. False

Which point is at (6, 2)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



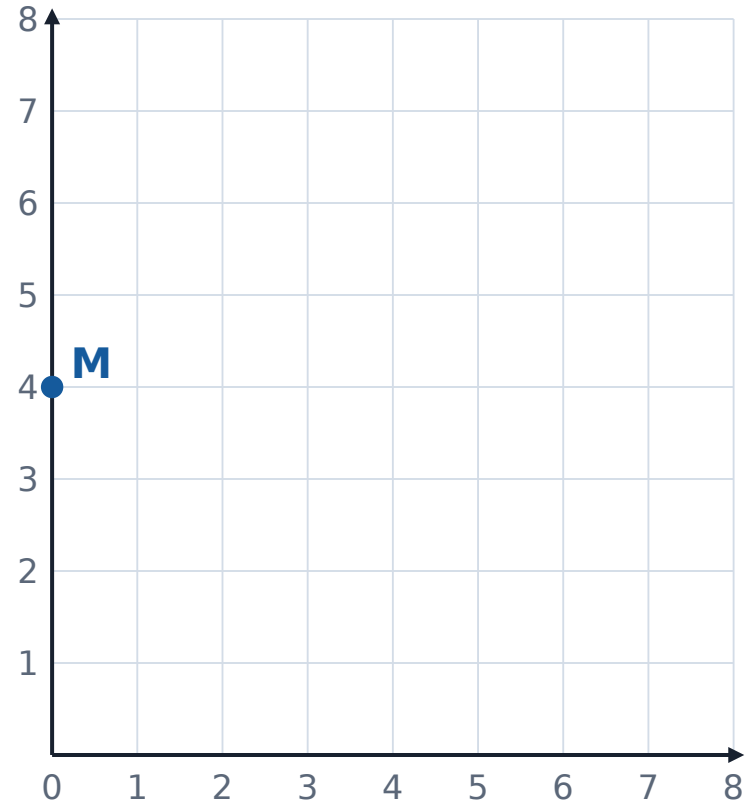
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 53 • BACKWARD 1 • Q3/5

What are the coordinates of point M?

M = _____



True or false?

The point $(5, 0)$ is on the y -axis.

A. True

B. False

SET 53 • 5.G.A.2

BACKWARD 2

Reading the scale on an axis • Grade 3 • Measurement & Data • 3.MD.B.3

An axis is numbered by 5s: 0, 5, 10, 15, ...

What value is 4 grid lines above 0?

- A.** 15
- B.** 25
- C.** 20
- D.** 4

True or false?

On an axis numbered by 2s, the value 7 is halfway between 6 and 8.

- A. True
- B. False

An axis is numbered by 10s.

What value is at the 3rd grid line above 0?

Value: _____

An axis is numbered by 3s.

Where is the value 12?

- A.** At the 4th grid line
- B.** At the 3rd grid line
- C.** Between the 3rd and 4th grid lines
- D.** At the 12th grid line

True or false?

On an axis numbered by 5s, the value 12 is on a grid line.

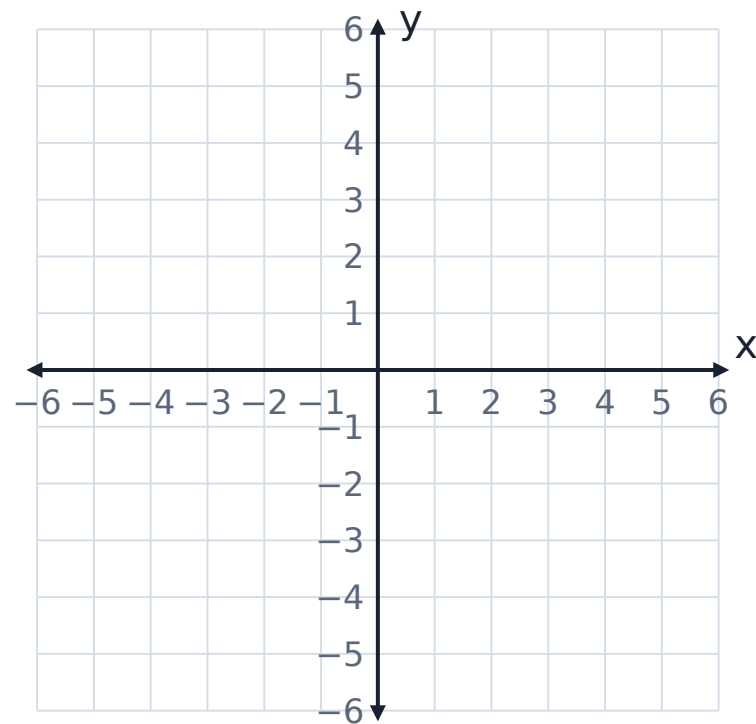
- A.** True
- B.** False

SET 53 • 5.G.A.2

FORWARD 1

Plot and reflect points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.b

Plot and label the points
 $P(-3, 4)$ and $Q(3, -4)$.



The point $(5, 2)$ is reflected across the x -axis.
What are the coordinates of its image?

Image: _____

In which quadrant is the point $(-2, -6)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

The point $(-4, 1)$ is reflected across the y -axis.
What are the coordinates of its image?

Image: _____

True or false?

The points $(3, 5)$ and $(-3, 5)$ are reflections of each other across the x-axis.

A. True

B. False

SET 53 • 5.G.A.2

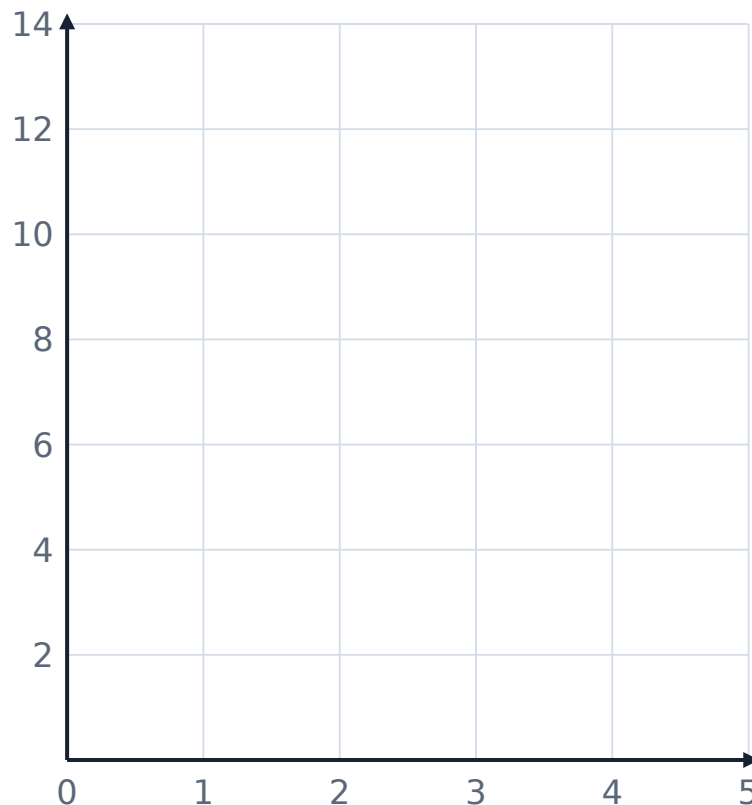
FORWARD 2

Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs (1, 3), (2, 6),
and (4, 12).

Is the relationship
proportional?

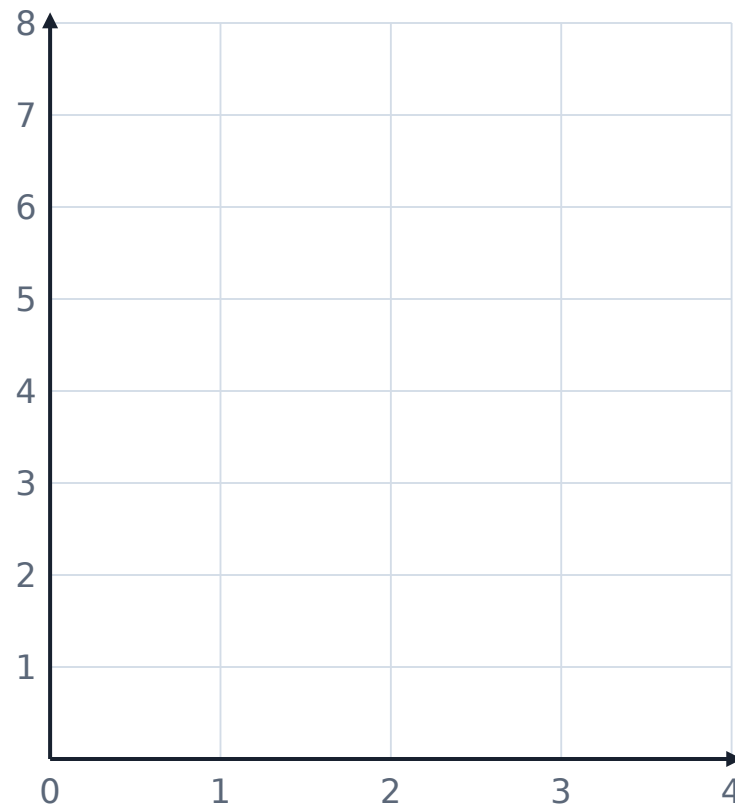
Answer: _____



Graph the pairs $(0, 2)$, $(1, 4)$,
and $(2, 6)$.

Is the relationship
proportional?

Answer: _____



Which graph shows a proportional relationship?

- A.** A curve through $(0, 0)$
- B.** Points that are not on a line
- C.** A straight line through $(0, 3)$
- D.** A straight line through $(0, 0)$

True or false?

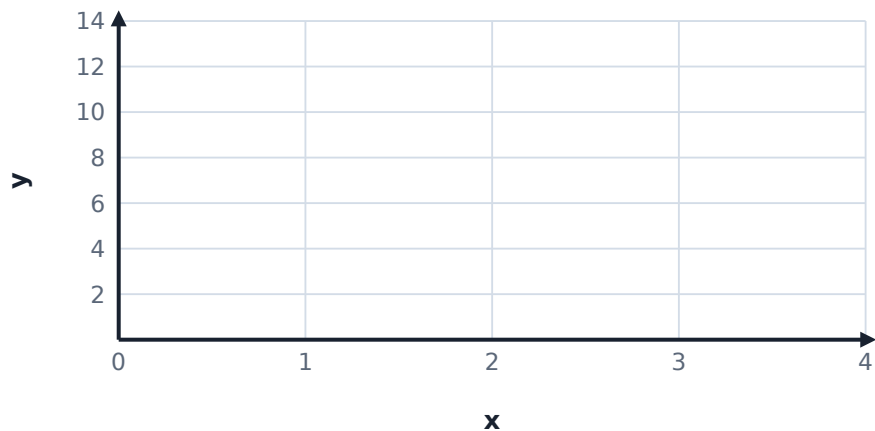
The points (2, 1), (4, 2), and (6, 3) show a proportional relationship.

- A. True
- B. False

Graph the pairs from the table.

Is the relationship proportional?

x	1	2	3
y	4	8	12



Answer: _____

Solve real-world problems by graphing points and interpreting coordinates

MAIN	Grade 5 main question	5.G.A.2
BACKWARD 1	The coordinate system	5.G.A.1
BACKWARD 2	Distance between whole numbers on a number line	2.MD.B.6
FORWARD 1	Find distances in all four quadrants	6.NS.C.8
FORWARD 2	Use a scale on a map grid	7.G.A.1

SET 54 • 5.G.A.2

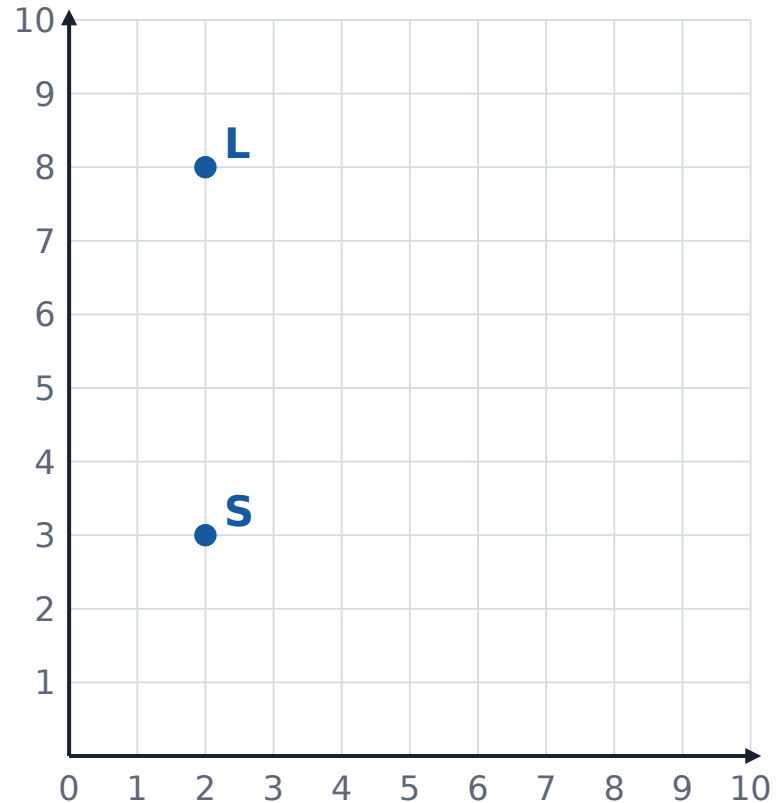
MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

The grid is a map. Each unit is 1 block.

How many blocks is it from the school S to the library L?

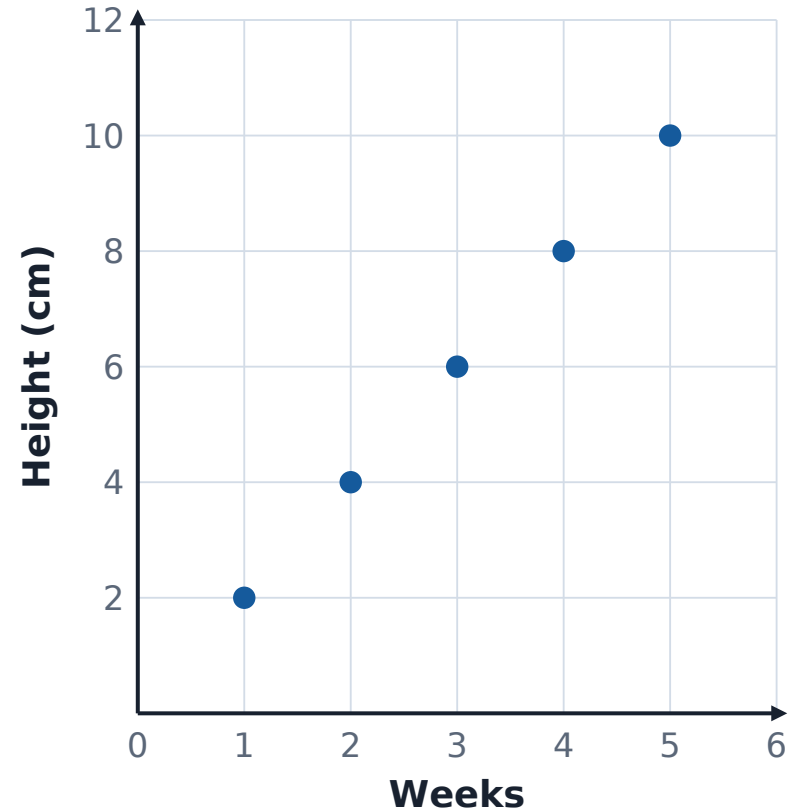
Blocks: _____



The graph shows the height of a plant.

How tall was the plant after 4 weeks?

Height: _____



The point (3, 12) is on a graph of hours worked (x) and dollars earned (y).

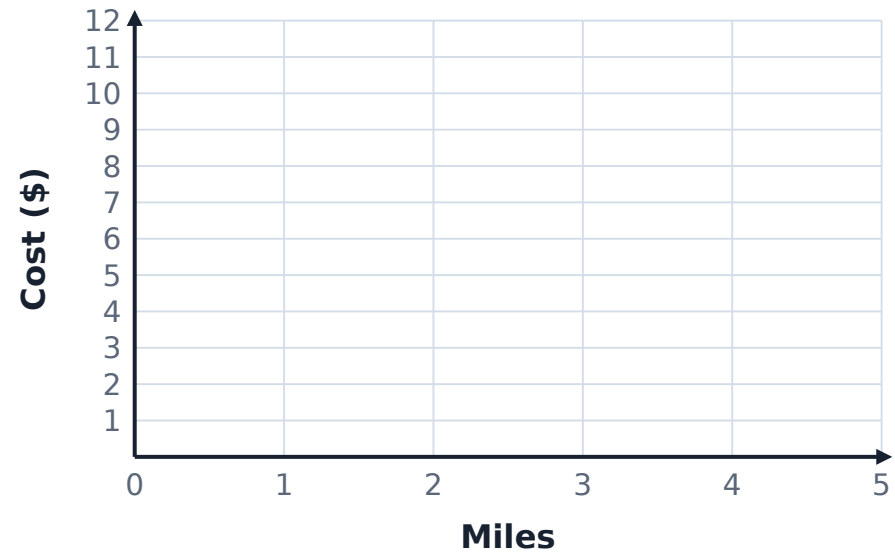
What does the point mean?

- A.** Working 3 hours earns \$12.
- B.** Working 12 hours earns \$3.
- C.** The pay starts at \$12.
- D.** Each hour earns \$3.

The table shows the cost of a taxi ride.

Graph the points.

Miles	1	2	3	4
Cost (\$)	5	7	9	11



On a map grid, the park is at (1, 6) and the pool is at (7, 6). Each unit is 100 meters.

How far apart are the park and the pool?

Distance: _____

SET 54 • 5.G.A.2

BACKWARD 1

The coordinate system • Grade 5 • Geometry • 5.G.A.1

What are the coordinates of the origin?

Origin: _____

True or false?

The first number in an ordered pair tells how far to move along the x-axis.

A. True

B. False

Point R is 4 units to the right of the origin and 1 unit up.
What are its coordinates?

A. (5, 0)

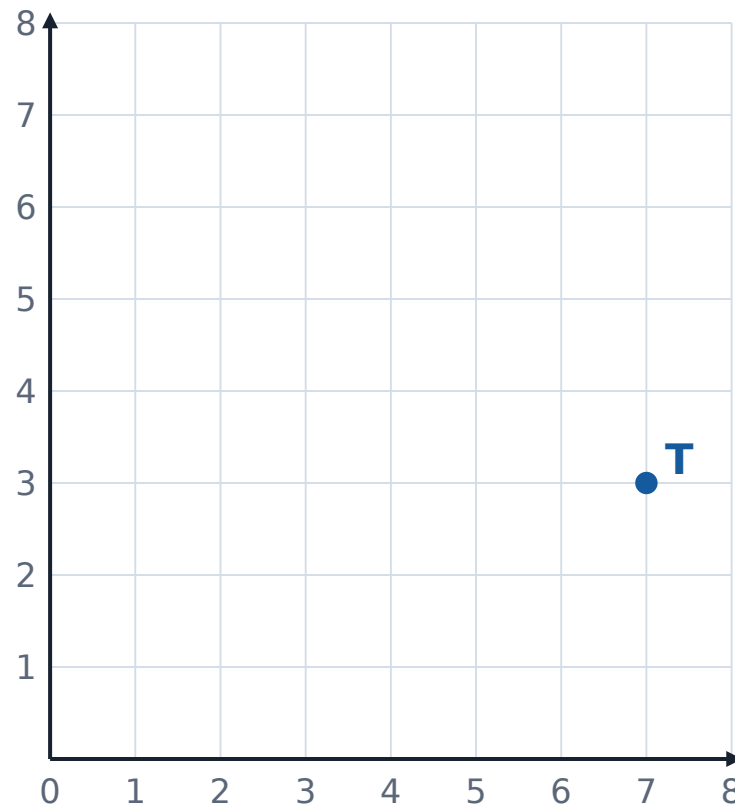
B. (1, 4)

C. (4, 4)

D. (4, 1)

What are the coordinates of point T?

T = _____



True or false?

$(0, 2)$ and $(2, 0)$ name the same point.

A. True

B. False

SET 54 • 5.G.A.2

BACKWARD 2

Distance between whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 2 to 7 on a number line?

Distance: _____

True or false?

The distance from 3 to 10 is 13 units.

A. True

B. False

How many units apart are 1 and 8 on a number line?

A. 9

B. 8

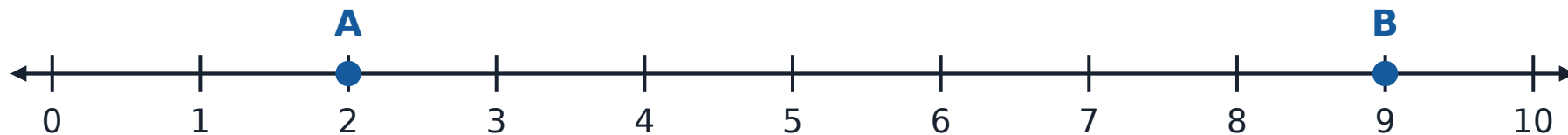
C. 1

D. 7

What is the distance from 6 to 6 on a number line?

Distance: _____

How many units apart are points A and B?



Units: _____

SET 54 • 5.G.A.2

FORWARD 1

Find distances in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-4, 2)$ and $(5, 2)$.

Distance: _____

Find the distance between $(3, -6)$ and $(3, 1)$.

Distance: _____

On a map, the store is at $(-2, -3)$ and the bank is at $(-2, 4)$. Each unit is 1 block.

How far apart are they?

A. 7 blocks

B. 1 block

C. 4 blocks

D. 6 blocks

A rectangle has vertices at $(-3, 1)$, $(2, 1)$, $(2, -2)$, and $(-3, -2)$.

What is its perimeter?

Perimeter: _____

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 0.

A. True

B. False

SET 54 • 5.G.A.2

FORWARD 2

Use a scale on a map grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is 2 miles. Two towns are at (1, 3) and (6, 3).

What is the actual distance between them?

Distance: _____

On a floor plan, each unit is $1\frac{1}{2}$ feet. A wall goes from (0, 0) to (0, 8).

How long is the real wall?

Length: _____

A scale drawing uses 1 grid unit = 5 m. A path on the grid is 7 units long.

How long is the real path?

A. 12 m

B. 35 m

C. 2 m

D. 7 m

A garden plan has corners at $(0, 0)$, $(6, 0)$, $(6, 4)$, and $(0, 4)$. Each unit is 3 feet.

What are the real length and width?

Length and width: _____

True or false?

If each grid unit is 10 km, the points (2, 5) and (2, 9) are 4 km apart.

- A. True
- B. False

Attributes of a category belong to all of its subcategories

MAIN	Grade 5 main question	5.G.B.3
BACKWARD 1	Quadrilaterals and their attributes	3.G.A.1
BACKWARD 2	Parallel lines, perpendicular lines, and angles	4.G.A.1
BACKWARD 3	Classifying shapes by sides and angles	4.G.A.2
FORWARD 1	Draw polygons in the coordinate plane	6.G.A.3
FORWARD 2	Decide whether shapes with given conditions are possible	7.G.A.2

SET 55 • 5.G.B.3

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.3

True or false?

All rectangles have four right angles. Squares are rectangles, so all squares have four right angles.

A. True

B. False

All parallelograms have two pairs of parallel sides.

Which shape must also have two pairs of parallel sides?

- A.** Kite
- B.** Pentagon
- C.** Triangle
- D.** Rhombus

A rhombus is a quadrilateral with 4 sides of equal length.

Is every square a rhombus? Explain.

Answer: _____

Explanation: _____

True or false?

All quadrilaterals have 4 right angles.

A. True

B. False

Which statement is true?

- A.** Every parallelogram is a square.
- B.** Every square is a rectangle.
- C.** Every quadrilateral is a rectangle.
- D.** Every rectangle is a square.

SET 55 • 5.G.B.3

BACKWARD 1

Quadrilaterals and their attributes • Grade 3 • Geometry • 3.G.A.1

How many sides does every quadrilateral have?

Sides: _____

True or false?

Triangles, rectangles, and squares are all quadrilaterals.

A. True

B. False

Which shape is NOT a quadrilateral?

- A.** Rectangle
- B.** Square
- C.** Rhombus
- D.** Triangle

True or false?

A square has 4 sides of equal length.

A. True

B. False

Name a quadrilateral that always has 4 right angles.

Shape: _____

SET 55 • 5.G.B.3

BACKWARD 2

Parallel lines, perpendicular lines, and angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Parallel lines are always the same distance apart.

A. True

B. False

Which angle measure is a right angle?

- A.** 90°
- B.** 180°
- C.** 45°
- D.** 120°

What do we call two lines that meet to form right angles?

Answer: _____

True or false?

An obtuse angle is smaller than a right angle.

A. True

B. False

How many pairs of parallel sides does a rectangle have?

A. 4

B. 0

C. 2

D. 1

SET 55 • 5.G.B.3

BACKWARD 3

Classifying shapes by sides and angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A shape with no parallel sides can still be a quadrilateral.

A. True

B. False

Which kind of triangle has a right angle?

- A.** An equilateral triangle
- B.** An acute triangle
- C.** A right triangle
- D.** An obtuse triangle

How many right angles does a square have?

Right angles: _____

True or false?

A rhombus always has 4 right angles.

A. True

B. False

Which shape always has two pairs of parallel sides?

A. Rectangle

B. Kite

C. Triangle

D. Pentagon

SET 55 • 5.G.B.3

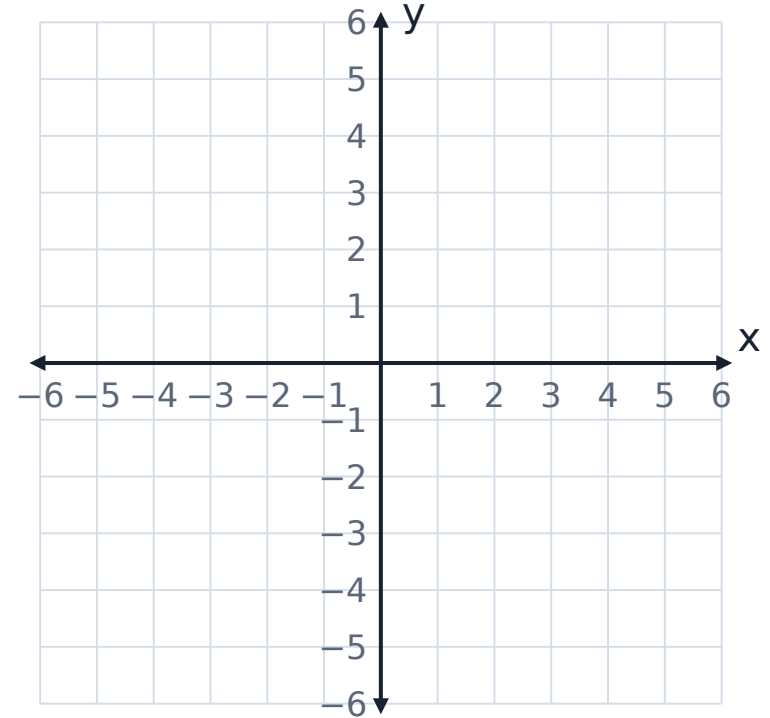
FORWARD 1

Draw polygons in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Plot $A(-2, 1)$, $B(3, 1)$, $C(3, -3)$, and $D(-2, -3)$. Connect them in order.

What polygon did you draw?

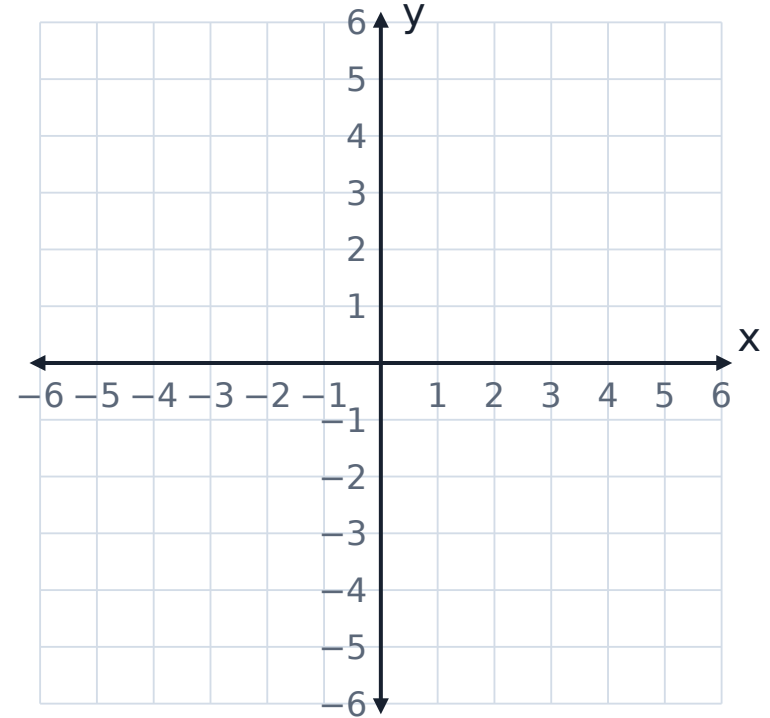
Polygon: _____



Plot $E(0, 3)$, $F(3, 0)$, $G(0, -3)$, and $H(-3, 0)$. Connect them in order.

What polygon did you draw?

Polygon: _____



The points $(-1, 2)$, $(4, 2)$, and $(4, -1)$ are three vertices of a rectangle.

What is the fourth vertex?

A. $(1, -1)$

B. $(-1, -1)$

C. $(-1, 4)$

D. $(4, 2)$

What is the length of the side from $(-2, 5)$ to $(4, 5)$?

Length: _____

True or false?

The points $(0, 0)$, $(4, 0)$, $(4, 2)$, and $(0, 2)$ are the vertices of a square.

- A. True
- B. False

SET 55 • 5.G.B.3

FORWARD 2

Decide whether shapes with given conditions are possible • Grade 7 •
Geometry • 7.G.A.2

True or false?

A triangle can have side lengths of 3 cm, 4 cm, and 5 cm.

A. True

B. False

Which three lengths can form a triangle?

A. 4, 5, 7

B. 5, 5, 10

C. 1, 1, 3

D. 2, 3, 6

Can a triangle have side lengths of 2 cm, 2 cm, and 5 cm? Write yes or no.

Answer: _____

True or false?

A triangle can have two right angles.

A. True

B. False

Can a quadrilateral have exactly 3 right angles? Write yes or no.

Answer: _____

Classify quadrilaterals in a hierarchy based on their properties

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Quadrilaterals	3.G.A.1
BACKWARD 2	Classifying by parallel sides and right angles	4.G.A.2
FORWARD 1	Area of parallelograms, rhombuses, and trapezoids	6.G.A.1
FORWARD 2	Shapes with given conditions	7.G.A.2

SET 56 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Square, rectangle, parallelogram, quadrilateral
- B.** Rectangle, square, quadrilateral, parallelogram
- C.** Quadrilateral, parallelogram, rectangle, square
- D.** Parallelogram, quadrilateral, square, rectangle

True or false?

A square is a rectangle, a rhombus, and a parallelogram.

A. True

B. False

Which of these categories does every rhombus belong to?

quadrilateral, parallelogram, rectangle, square

Categories: _____

A shape has 4 sides of equal length and 4 right angles.
What is the most specific name for it?

- A.** Rectangle
- B.** Rhombus
- C.** Square
- D.** Quadrilateral

True or false?

Every parallelogram is a rhombus.

A. True

B. False

SET 56 • 5.G.B.4

BACKWARD 1

Quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle is a quadrilateral.

A. True

B. False

Which shape always has 4 sides of equal length?

- A.** Triangle
- B.** Rectangle
- C.** Rhombus
- D.** Kite

How many angles does a quadrilateral have?

Angles: _____

True or false?

Every quadrilateral is a square.

A. True

B. False

Name a shape that is a quadrilateral but is not a parallelogram.

Shape: _____

SET 56 • 5.G.B.4

BACKWARD 2

Classifying by parallel sides and right angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A quadrilateral with 2 pairs of parallel sides is a parallelogram.

A. True

B. False

Which shape always has 4 right angles?

- A.** Rectangle
- B.** Parallelogram
- C.** Kite
- D.** Rhombus

How many pairs of parallel sides does a parallelogram have?

Pairs: _____

True or false?

The sides of a square that meet at a corner are perpendicular.

- A. True
- B. False

Which description fits every rhombus?

- A.** No parallel sides
- B.** 3 sides
- C.** 4 sides of equal length
- D.** 4 right angles

SET 56 • 5.G.B.4

FORWARD 1

Area of parallelograms, rhombuses, and trapezoids • Grade 6 • Geometry •
6.G.A.1

A parallelogram has a base of 10 cm and a height of 6 cm.

What is its area?

Area: _____

A rhombus has a base of 7 in. and a height of 5 in.
What is its area?

Area: _____

A square has sides that are 9 m long.

What is its area?

- A.** 36 square meters
- B.** 81 square meters
- C.** 90 square meters
- D.** 18 square meters

A trapezoid has bases of 6 ft and 10 ft and a height of 4 ft.

What is its area?

Area: _____

True or false?

A parallelogram has a greater area than a rectangle with the same base and the same height.

A. True

B. False

SET 56 • 5.G.B.4

FORWARD 2

Shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Can you draw a rectangle that is not a square? Write yes or no.

Answer: _____

A quadrilateral has 4 sides of equal length but no right angles.

What is it?

- A.** A rhombus that is not a square
- B.** A trapezoid
- C.** A rectangle
- D.** A square

True or false?

Only one triangle can be drawn with angles of 40° , 60° , and 80° .

A. True

B. False

Is there more than one triangle shape with sides of 6 cm, 6 cm, and 6 cm? Write yes or no.

Answer: _____

True or false?

A triangle with three sides of 5 cm also has three equal angles.

- A. True
- B. False

Classify triangles by their sides and angles

MAIN	Grade 5 main question	5.G.B.4
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MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

A triangle has sides of 6 cm, 6 cm, and 6 cm.

Which name fits it?

- A.** Obtuse
- B.** Scalene
- C.** Equilateral
- D.** Right

A triangle has angles of 90° , 50° , and 40° .
Classify it by its angles.

Classification: _____

True or false?

Every equilateral triangle is also an isosceles triangle.

A. True

B. False

A triangle has sides of 3, 5, and 7 units and an angle of 120° .

Which name fits it?

A. Acute equilateral

B. Right scalene

C. Right isosceles

D. Obtuse scalene

A triangle has angles of 70° , 70° , and 40° .
Classify it by its angles and by its sides.

By angles: _____

By sides: _____

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BACKWARD 1

Right, acute, and obtuse triangles • Grade 4 • Geometry • 4.G.A.2

True or false?

A right triangle has exactly one right angle.

A. True

B. False

Which angle measure is obtuse?

- A.** 10°
- B.** 45°
- C.** 130°
- D.** 90°

A triangle has one angle greater than 90° .
What kind of triangle is it?

Kind: _____

True or false?

In an acute triangle, all three angles are less than 90° .

A. True

B. False

Which angle measure is acute?

A. 100°

B. 180°

C. 60°

D. 90°

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BACKWARD 2

Kinds of angles • Grade 4 • Geometry • 4.G.A.1

True or false?

A right angle measures 90° .

A. True

B. False

What kind of angle measures 35° ?

- A.** Right
- B.** Straight
- C.** Obtuse
- D.** Acute

What kind of angle measures 150° ?

Kind: _____

True or false?

Two right angles together make 360° .

A. True

B. False

What kind of angle measures exactly 90° ?

Kind: _____

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BACKWARD 3

Triangles • Grade 2 • Geometry • 2.G.A.1

How many sides does a triangle have?

Sides: _____

True or false?

A triangle has 3 angles.

A. True

B. False

Which shape has 3 sides?

- A.** Square
- B.** Hexagon
- C.** Pentagon
- D.** Triangle

True or false?

All triangles have sides of equal length.

A. True

B. False

How many vertices (corners) does a triangle have?

Vertices: _____

SET 57 • 5.G.B.4

FORWARD 1

Area of triangles • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs that are 6 cm and 8 cm long.
What is its area?

Area: _____

A triangle has a base of 10 in. and a height of 7 in.
What is its area?

Area: _____

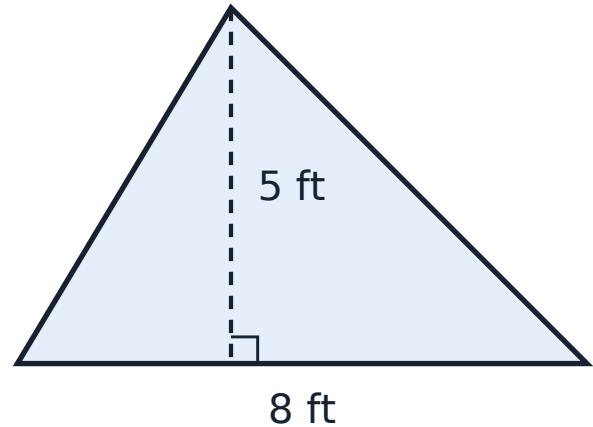
An obtuse triangle has a base of 5 m. Its height, measured outside the triangle, is 4 m.

What is its area?

- A.** 20 square meters
- B.** 40 square meters
- C.** 10 square meters
- D.** 9 square meters

Find the area of the triangle.

Area: _____



True or false?

Two triangles with the same base and the same height have the same area.

A. True

B. False

SET 57 • 5.G.B.4

FORWARD 2

Complementary, supplementary, and vertical angles • Grade 7 • Geometry •
7.G.B.5

Two angles are complementary. One measures 35° .
What is the measure of the other angle?

Angle: _____

Two angles are supplementary. One measures 110° .
What is the measure of the other angle?

Angle: _____

Vertical angles are always ____.

- A.** right angles
- B.** complementary
- C.** equal
- D.** supplementary

An angle and a 72° angle together form a straight line.
What is the measure of the angle?

Angle: _____

True or false?

Two adjacent angles that together form a right angle are complementary.

- A. True
- B. False